

A ADDITIONAL PROOFS OF LEMMAS AND THEOREMS

LEMMA 4.1. Given a graph G and two h -clique λ -compact subgraphs $G[X]$ and $G[Y]$, if $X \cap Y \neq \emptyset$ or there exists a cross h -clique between them (i.e., $\Psi_h(X, Y) \neq \emptyset$), then $G[X \cup Y]$ is also h -clique λ -compact subgraph of G .

PROOF. If $X \cap Y \neq \emptyset$ or $\Psi_h(X, Y) \neq \emptyset$, then the induced subgraph $G[X \cup Y]$ is connected. Let $S \subseteq X \cup Y$ be any non-empty vertex subset to be removed, and define

$$\Delta\Psi_h(G[S]) = \{C \in \Psi_h(G[X \cup Y]) \mid C \cap S \neq \emptyset\}$$

as the set of h -cliques that disappear after deleting S .

We partition S into two disjoint parts:

$$S_X = S \cap X, \quad S_Y = S \cap (Y \setminus X).$$

Each h -clique in $\Psi_h(G[X])$ that contains a vertex from S_X must also belong to $\Psi_h(G[X \cup Y])$, and the same holds for $G[Y]$. Therefore, the number of h -cliques in $G[X \cup Y]$ removed by deleting S_X (or S_Y) is at least the number removed in $G[X]$ (or $G[Y]$), i.e.,

$|\{C \in \Psi_h(G[X \cup Y]) \mid C \cap S_X \neq \emptyset\}| \geq |\{C \in \Psi_h(G[X]) \mid C \cap S_X \neq \emptyset\}|$,
 $|\{C \in \Psi_h(G[X \cup Y]) \mid C \cap S_Y \neq \emptyset\}| \geq |\{C \in \Psi_h(G[Y]) \mid C \cap S_Y \neq \emptyset\}|$.
 Since both X and Y are h -clique λ -compact, we have $|\{C \in \Psi_h(G[X]) \mid C \cap S_X \neq \emptyset\}| \geq \lambda|S_X|$, and $|\{C \in \Psi_h(G[Y]) \mid C \cap S_Y \neq \emptyset\}| \geq \lambda|S_Y|$.
 Hence, the total number of h -cliques in $\Psi_h(G[X \cup Y])$ that are removed by deleting S satisfies

$$|\Delta\Psi_h(G[S])| \geq \lambda(|S_X| + |S_Y|) = \lambda|S|.$$

Thus, $G[X \cup Y]$ satisfies the condition of the h -clique λ -compact subgraph. $\square$

THEOREM 4.4. A maximal h -clique λ -compact subgraph $\mathcal{H}$ is an LhCDS if and only if it contains no h -clique λ' -compact subgraph with $\lambda' > \eta_h(\mathcal{H})$.

PROOF. To prove this theorem, we need two facts.

Fact 1. For any connected subgraph H , $\eta_h(H) \leq \rho_h(H)$. *Proof.* Since H is h -clique $\eta_h(H)$ -compact, deleting $V(H)$ removes $|\Psi_h(H)|$ cliques. Thus $|\Psi_h(H)| \geq \eta_h(H)|V(H)|$, which implies $\rho_h(H) \geq \eta_h(H)$.

Fact 2. For any connected subgraph H , $\eta_h(H) < \rho_h(H)$ iff there exists $H' \subset H$ with $\rho_h(H') > \rho_h(H)$. *Proof.* ($\Rightarrow$) If $\eta_h(H) < \rho_h(H)$, then H is not $\rho_h(H)$ -compact: some $S \subset V(H)$ deletes fewer than $\rho_h(H)|S|$ cliques. Let $H' = H \setminus S$. Then

$$|\Psi_h(H')| > |\Psi_h(H)| - \rho_h(H)|S| = \rho_h(H)(|V(H)| - |S|) = \rho_h(H)|V(H')|,$$

so $\rho_h(H') > \rho_h(H)$.

($\Leftarrow$) If $H' \subset H$ with $\rho_h(H') > \rho_h(H)$, let $S = V(H) \setminus V(H')$. Then

$$|\Psi_h(H)| - |\Psi_h(H')| < \rho_h(H)(|V(H)| - |V(H')|) = \rho_h(H)|S|,$$

so H is not $\rho_h(H)$ -compact, hence $\eta_h(H) < \rho_h(H)$.

Corollary. $\eta_h(H) = \rho_h(H)$ iff no proper subgraph of H has larger density.

($\Rightarrow$) If $\mathcal{H}$ is an LhCDS, then by the corollary $\eta_h(\mathcal{H}) = \rho_h(\mathcal{H})$. If $\mathcal{H}$ contained an h -clique λ' -compact subgraph $\mathcal{H}'$ with $\lambda' > \eta_h(\mathcal{H})$, then

$$\rho_h(\mathcal{H}') \geq \eta_h(\mathcal{H}') \geq \lambda' > \eta_h(\mathcal{H}) = \rho_h(\mathcal{H}),$$

contradiction.

($\Leftarrow$) Suppose $\mathcal{H}$ is maximal h -clique λ -compact but not an LhCDS. Then:

(a) If $\mathcal{H}$ has a denser proper subgraph $\mathcal{H}'$, take a densest subgraph $\mathcal{H}'' \subseteq \mathcal{H}'$. Then

$$\eta_h(\mathcal{H}'') = \rho_h(\mathcal{H}'') > \rho_h(\mathcal{H}) \geq \eta_h(\mathcal{H}),$$

so $\mathcal{H}''$ is h -clique λ' -compact with $\lambda' > \eta_h(\mathcal{H})$.

(b) If $\mathcal{H}$ is locally dense but not maximal, there exists $\widehat{\mathcal{H}} \supset \mathcal{H}$ with $\eta_h(\widehat{\mathcal{H}}) = \rho_h(\widehat{\mathcal{H}}) \geq \eta_h(\mathcal{H}) = \lambda$, contradicting maximal λ -compactness. Hence this case is impossible.

Thus only case (a) can occur, and $\mathcal{H}$ must contain an h -clique λ' -compact subgraph with $\lambda' > \eta_h(\mathcal{H})$. $\square$

LEMMA 4.13. Let $\{B_i\}_{i=0}^r$ be the sequence of locally h -clique dense subgraphs. For any two indices i and j such that $0 \leq i < j \leq r$, let $B_l = F_h(\lambda)$ with $\lambda = \rho_h(B_j, B_i)$. Then:

- If $i + 1 < j$, then $i < l < j$.
- If $i + 1 = j$, then $l = j$.

PROOF. Let the locally dense subgraphs satisfy $\emptyset = B_0 \subset B_1 \subset \dots \subset B_r = G$. By corollary 4.8, the outer densities satisfy $\rho_h(B_k, B_{k-1}) > \rho_h(B_{k+1}, B_k)$ for all $0 < k < r$. Let $\rho_h(B_k, B_{k-1}) = \frac{c_k}{v_k}$, where $c_k = |\Psi_h(G[B_k])| - |\Psi_h(G[B_{k-1}])|$ and $v_k = |B_k| - |B_{k-1}|$. Then

$$\frac{c_{i+1}}{v_{i+1}} > \frac{c_{i+2}}{v_{i+2}} > \dots > \frac{c_j}{v_j}.$$

The outer density between B_i and B_j is

$$\rho_h(B_j, B_i) = \frac{\sum_{k=i+1}^j c_k}{\sum_{k=i+1}^j v_k},$$

which lies strictly between $\frac{c_{i+1}}{v_{i+1}}$ and $\frac{c_j}{v_j}$. Thus,

$$\rho_h(B_j, B_{i+1}) < \rho_h(B_j, B_i) < \rho_h(B_{i+1}, B_i),$$

and by the definition of $F_h(\lambda)$, we have $i < l < j$. When $i + 1 = j$, we have $\lambda = \rho_h(B_j, B_i) = \rho_h(B_{i+1}, B_i)$, implying $l = j$. $\square$

THEOREM 5.6. Given an h -clique orientation graph $\vec{\mathcal{G}}_h$, the procedure Orientation-par correctly outputs the h -clique density-based partition Θ_h .

PROOF. We first characterize the structural condition under which a layer boundary is well-defined. For any integer $i \geq 0$, define

$$R_l = \{u \in V \mid d(u, \vec{\mathcal{G}}_h) < i - 1\}, \quad R_u = \{u \in V \mid d(u, \vec{\mathcal{G}}_h) > i - 1\}.$$

If there exists no directed path in $\vec{\mathcal{G}}_h$ from any vertex in R_u to any vertex in R_l , then the orientation is *balanced* at threshold $i - 1$, and the i -th layer satisfies

$$\mathcal{R}_i = R_u \cup \{u \in V \mid u \text{ can reach some vertex in } R_u\}.$$

This provides the structural foundation of layer separation by the Theorem A.1.

THEOREM A.1 (LAYER CHARACTERIZATION). Given an h -clique orientation graph $\vec{\mathcal{G}}_h$ and an integer $i \geq 0$, let

$$R_l \triangleq \{u \in V \mid d(u, \vec{\mathcal{G}}_h) < i - 1\}, \quad R_u \triangleq \{u \in V \mid d(u, \vec{\mathcal{G}}_h) > i - 1\}.$$

If there is no directed path in $\vec{\mathcal{G}}_h$ from any vertex in R_u to any vertex in R_l , then

$$\mathcal{R}_i = \mathcal{R}_u \cup \{u \in V \mid u \text{ can reach some vertex in } \mathcal{R}_u\}.$$

Table 6: Descriptions of the different variants.

Name	Divide-and-conquer	Clique orientation partition
IPPV	✗	✗
IPPV+Par	✗	✓
Divide	✓	✗
DCLDS	✓	✓

PROOF. Let $T \triangleq R_u \cup \{u \in V \mid u \text{ can reach a vertex in } R_u\}$. Based on the definition of $\mathcal{R}_i$, it suffices to prove $T = \mathcal{R}_i$. There are no paths from R_l to R_u implies the orientation is *balanced* at threshold $i - 1$: no vertex with in-degree $< i - 1$ can forward flow to one with in-degree $> i - 1$. Consequently, all cross- h -cliques between T and $V \setminus T$ are oriented away from T , and every vertex in T has $d(u, \tilde{\mathcal{G}}_h) \geq i - 1$. $\square$

Next, the procedure Flow constructs the reorientation network $\mathcal{F}_h^{i-1}$ and computes the maximum flow between sources and sinks representing R_l and R_u . By the max-flow min-cut theorem, the flow is maximum if and only if no s - t augmenting path remains. Each such path corresponds to a directed path from R_l to R_u in $\tilde{\mathcal{G}}_h$ (Definition 5.5). Thus, after computing the max flow, the residual orientation satisfies the balanced condition above, ensuring that the vertex set returned by Flow is exactly $\mathcal{R}_i$.

Finally, Orientation-par recursively invokes Flow over integer thresholds to identify all partition layers. By induction on recursion depth, each computed $\mathcal{R}_i$ is correct, and every possible threshold is explored exactly once. When adjacent indices differ by one or yield identical layers, recursion terminates, guaranteeing that all disjoint layers are discovered. Therefore, the union of all $\mathcal{R}_i$ constitutes the complete and exact h -clique density-based partition Θ_h . $\square$

1. Time cost of different steps in DCLDS. Recall that DCLDS sequentially performs the following three steps: (1) enumerating all the h -cliques from the graph (CliqueEnum), (2) partitioning the graph into small subgraphs Orientation-Par, (3) discovering the top- k LhCDSes via the divide-and-conquer method DeriveLhCDS. Figure 14 shows the time cost of these three steps on ten datasets, where $k = 10$, $h = 3$, and the graphs are assumed to be loaded into memory. We see that DeriveLhCDS is the most computationally expensive step on most of the graphs.

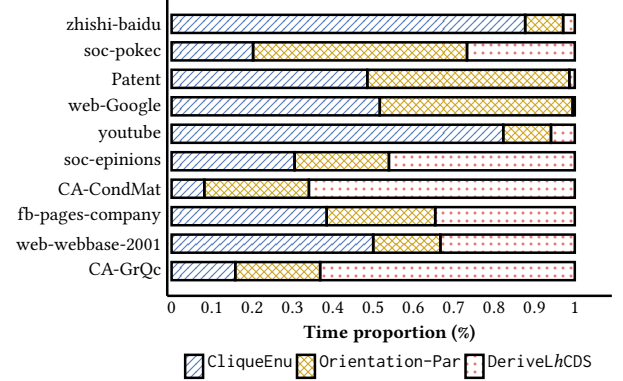


Figure 14: Proportion of time cost of each step in DCLDS.

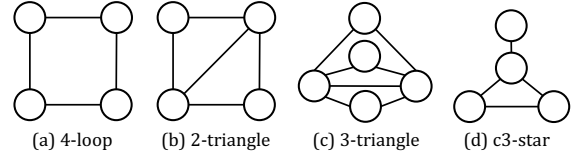


Figure 15: An example of four patterns.